Twitter Sentiment Analysis

Understanding Consumer Sentiment for Apple & Google Products

Daniella | Eve | Verah | Naomi

SXSW Tweet Dataset • April 2026

NLP | Machine Learning | Social Media | Brand Analytics

Project Overview

Business Problem: Companies receive millions of social media mentions daily. Marketing teams cannot manually read every tweet to understand customer sentiment.

The Solution: Build an NLP pipeline that automatically classifies tweet sentiment, transforming unstructured text into actionable brand intelligence.

The impact would be:

📡 Real-time Monitoring

Detect shifts in public sentiment toward Apple & Google as they happen at SXSW.

⚠️ Early Issue Detection

Surface negative feedback about products before it escalates or goes viral.

📊 Launch Evaluation

Quantify sentiment impact of product announcements with measurable metrics.

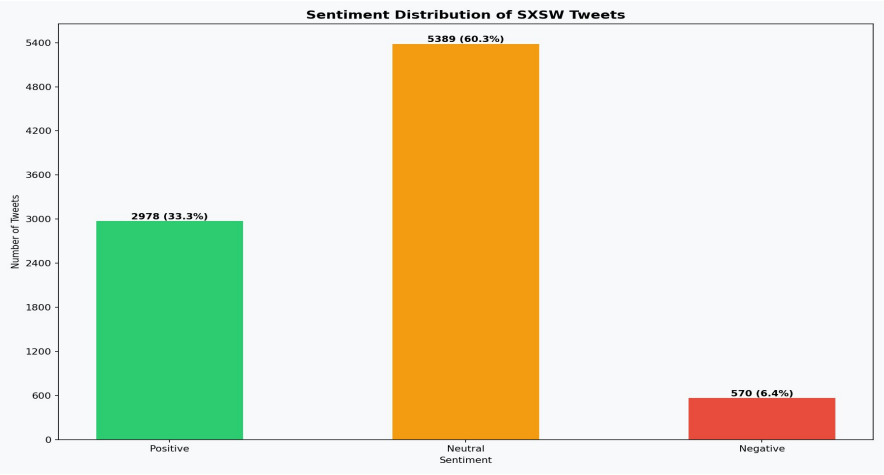
🔍 Competitive Insights

Compare brand perception between Apple and Google across tweet streams.

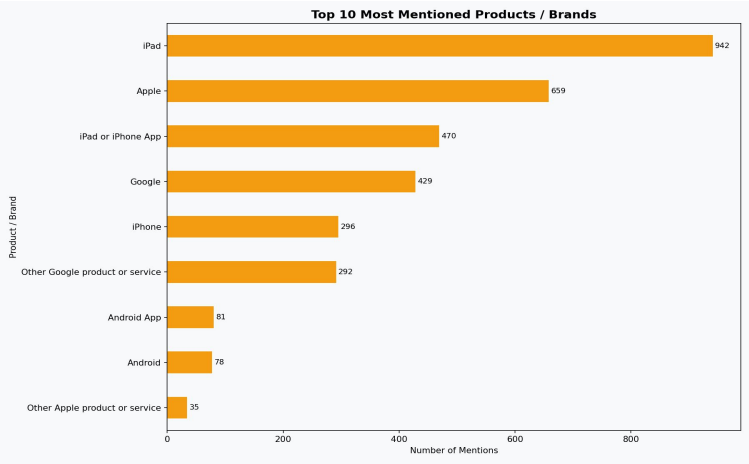
Dataset & Exploratory Analysis

9,093	8,937	3	~6%
Total Tweets	After Filtering	Sentiment Classes	Negative Tweets

Sentiment Distribution



Top Product Categories



Key Insight:

Severe class imbalance- Negative tweets are only 6% of the data. This drove our choice of weighted F1 as the primary evaluation metric over accuracy.

Apple products dominate the conversation (80% of mentions), with apps being the single largest category. This reflects SXSW's tech-conference context.

NLP Pipeline

A 3-stage pipeline transforms raw tweets into model-ready features

01 Text Cleaning

- → Lowercase normalisation
- → Remove URLs & @mentions
- → Strip # symbols (keep words)
- → Remove numbers & punctuation
- → Remove stopwords

02 TF-IDF Vectorisation

- → Max 5,000 features - most important features (words + bigrams)
- → Unigrams + Bigrams (1,2)
- → Bigrams capture phrases like "not good" (different meaning than "good")
- → Fit on training data only
- → Prevents data leakage

03 Label Encoding

- → Binary: Positive (1) / Negative (0)
- → Multiclass: Pos(2) / Neu(1) / Neg(0)
- → Stratified 80/20 train-test split
- → Preserves class proportions
- → Handles class imbalance fairly

Models Trained

Binary Classification

Positive vs. Negative • 3,548 tweets

✦ Logistic Regression

Interpretable baseline; coefficients reveal which words drive sentiment predictions. Uses L2 regularisation (C=1.0).

✦ Multinomial Naive Bayes

Probabilistic; models word likelihoods per class. Fast, robust to imbalance, strong on sparse TF-IDF matrices.

Multiclass Classification

Positive / Neutral / Negative • 8,937 tweets

✦ Random Forest

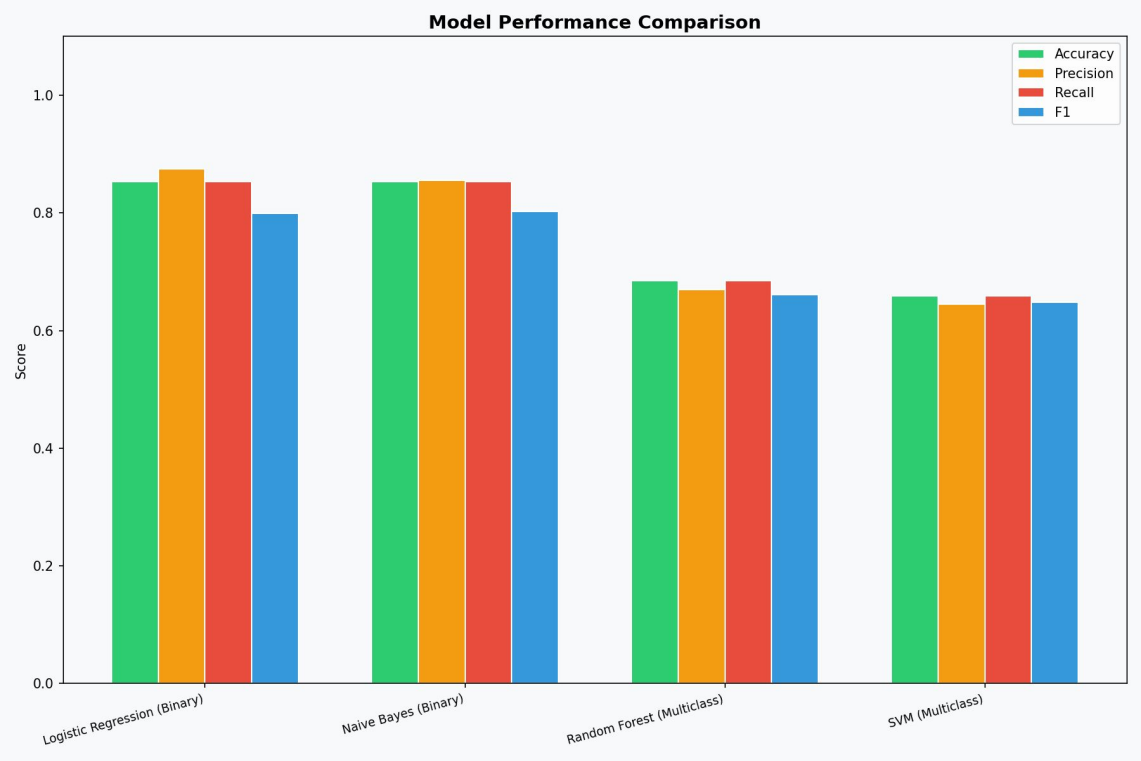
Ensemble of decision trees; handles non-linear boundaries. 100 estimators, max depth 10 to prevent overfitting.

✦ Linear SVM

High-dimensional text classifier; maximises margin between classes. Effective on large TF-IDF feature spaces.

Model Performance Results

Ranked by Weighted F1 Score (primary metric — robust to class imbalance)



🏆 **Best Model**

Naive Bayes (Binary)

F1 Score:	**0.8086** (Primary metric - handles class imbalance)
Accuracy:	**85.8%**
Precision:	**86.9%**

Key Insights

- Binary classifiers significantly outperform multiclass models — simpler task, cleaner signal
- Negative class hardest to detect — only ~6% of training data is negative
- SVM beats Random Forest on multiclass — linear models excel on sparse TF-IDF

Limitations & Caveats

📋 Data Limitations

✗ SXSW-specific context — may not generalise to non-tech events

✗ English tweets only — cannot analyse global audience

✗ Dated dataset — does not reflect current product/slang

✗ Human label subjectivity — single-rater annotations

🤖 Model Limitations

✗ No sarcasm detection — ironic tweets often misclassified

✗ Severe class imbalance — negative class underrepresented at 6%

✗ Neutral/Positive boundary is ambiguous even for humans

✗ No emoji handling — emojis stripped, losing sentiment signal

⚙️ Technical Limitations

✗ TF-IDF ignores word order — "not good" ≈ "good not"

✗ Fixed vocabulary — new slang goes out-of-vocabulary

✗ No conversational context — thread replies not analysed

✗ No transformer models — BERT/RoBERTa would perform better

Business Recommendations

📣 Marketing Teams

→ Track positive word frequency

Early indicator of campaign success — "love", "great", "awesome" are strong positive predictors.

→ Monitor launch windows

Compare pre/post sentiment to quantify the impact of announcements in real time.

→ Identify brand advocates

Users with consistent positive tweets are prime ambassador programme candidates.

🛠️ Product Teams

→ Prioritise negative tweet issues

"Crash", "battery", "slow" signal recurring complaints worth fast-tracking.

→ A/B test messaging with model

Use predicted sentiment scores to evaluate communication strategies before launch.

→ Expand training data

More negative examples (SMOTE or collection) will most improve recall on critical alerts.

Conclusions

01 Dataset is heavily skewed — Negative tweets are only ~6%, making F1 the right evaluation metric over accuracy.

02 Naive Bayes achieved the best F1 (0.81) on binary classification; SVM led multiclass (F1 = 0.67).

03 Negative sentiment detection is the hardest and most valuable class to improve — invest in more training data.

04 TF-IDF + unigrams/bigrams is a strong, interpretable baseline; transformer models (BERT) are the clear next step.